Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2025-P1-Model)** **Date: 22-03-2026**

Time: 09.00Am to 12.00Pm **GTA-19** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**C**	2	**D**	3	**B**	4	**C**	5	**AB**	6	**AB**
7	**BD**	8	**738**	9	**1.5**	10	**1.5**	11	**4.8**	12	**3.00**
13	**12**	14	**C**	15	**A**	16	**A**				

PHYSICS

17	**C**	18	**B**	19	**C**	20	**C**	21	**AC**	22	**ABD**
23	**ABCD**	24	**124 – 127**	25	**1.5**	26	**14.13-14.15**	27	**3**	28	**0.25**
29	**0.56 - 0.58**	30	**C**	31	**D**	32	**C**				

CHEMISTRY

33	**B**	34	**D**	35	**C**	36	C	37	**ABC**	38	**ABC**
39	**ABD**	40	**7**	41	**128 – 132**	42	45	43	**30**	44	**13**
45	**0.66 - 0.67**	46	**D**	47	**A**	48	D				

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SOLUTIONS

MATHEMATICS

1. $a\left[2\sin\frac{3x}{2}\cos\frac{x}{2}\right]=2\sin\frac{3x}{2}\cos\frac{3x}{2}$

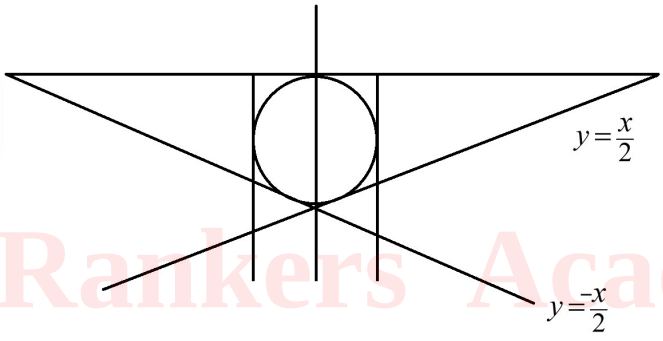
$\sin\frac{3x}{2}=0$ $\qquad a=4\cos^2\frac{x}{2}-3$

$\frac{3x}{2}=n\pi$ $\qquad a\in(-3,1)$

$x=\frac{2x\pi}{3}$ $\qquad a=-2,-1,0$

$x=\frac{2\pi}{3}$ $\qquad$ But for $a=-2$

$x=\frac{2\pi}{3}$

2. 

Centre $(0,\sqrt{5})$

3. $2^{\frac{1}{5}}\left(\frac{8}{3}\right)\le\int_0^2 x^2(x^2+2)^{\frac{1}{3}}dx\le 6^{\frac{1}{5}}\times\frac{8}{3}$

4. S = Coefficient of x^{10} in

$$\left\{{}^{10}C_1(1+x)^{20}-{}^{10}C_1(1+x)^{18}+{}^{10}C_2(1+x)^{16}-{}^{10}C_3(1+x)^{14}+{}^{10}C_4(1+x)^{12}-{}^{10}C_5(1+x)^{10}\right\}$$

= Coefficient of x^{10} in ${}^{10}C_0\left[(1+x)^2\right]^{10}-{}^{10}C_1\left[(1+x)^2\right]^9+{}^{10}C_2\left[(1+x)^2\right]^8$

$-{}^{10}C_3\left[(1+x)^2\right]^7+{}^{10}C_4\left[(1+x)^2\right]^6+{}^{10}C_5\left[(1+x)^2\right]^5+{}^{10}C_6\left[(1+x)^2\right]^4+____$

= Coefficient of x^{10} in $\left[(1+x)^2-1\right]^{10}$

5. $p=\frac{d}{dx}\left[\left(x^3+1\right)g\left(x^3\right)\right]$

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$p=\left(x^3+1\right)g'\left(x^3\right)3x^2+g\left(x^3\right)3x^2$

$p\left(7^{\frac{-2}{3}}\right)=24g'(7)+3g(7)$

None $g\left[f\left(x^5+x-2\right)\right]=g\left(x^3-1\right)$

$\therefore \quad g(7)=32, \quad g'(7)=\frac{27}{4}$

$\therefore \quad p\left(7^{\frac{-2}{3}}\right)=258$

Similarly $q=\frac{-3}{8}$.

6. $\int_0^{n+1}f(x)dx-\int_0^n f(x)dx=\int_n^{n+1}n^{x-n}dx=\frac{1}{n^n}\left[\frac{n^x}{\ln n}\right]_n^{n+1}=\frac{n-1}{\ln n}$

$\int_0^{n+1}g(x)dx-\int_0^n g(x)dx=\int_n^{n+1}(x-n)^n dx=\frac{(x-n)^{n+1}}{n+1}\Big]_n^{n+1}=\frac{1}{n+1}$

$t\in(0,1)\Rightarrow\int_0^t f(x)dx=0, t\in\left[1,\frac{\pi}{2}\right)\Rightarrow\int_0^t f(x)dx<\frac{\pi}{2}-1$

$t=1\Rightarrow\int_0^t g(x)dx=1$

$\lim_{n\to\infty}\int_0^n g(x)dx=\lim_{n\to\infty}\sum_{i=0}^{n}\frac{1}{i+1}\rightarrow$ Divergent series

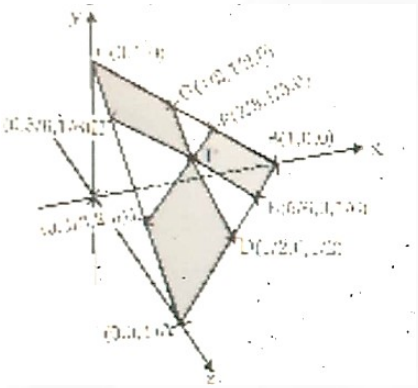
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7. $A(\alpha)$ will be area of ΔABC which is an equilateral

$\Delta A(\alpha)=\frac{\sqrt{3}}{4}(\sqrt{1+1})^2=\frac{\sqrt{3}}{2}$ sq. unit for $A(\beta)$, consider the point $P\left(\frac{1}{2},\frac{1}{3},\frac{1}{6}\right)$ in the plane $x+y+z=1$

The region β is shown as shaded region



$\Delta PFG, PDE$ and PHI are equilateral triangles and we find required area by removing their total area from $\text{ar}(\Delta ABC)$ Now FI is line through P parallel to AB

Equation of FI : $\frac{x-\frac{1}{2}}{1}=\frac{y-\frac{1}{3}}{0}=\frac{z-\frac{1}{6}}{-1}$

$$F\left(\frac{2}{3},\frac{1}{3},0\right)$$

at $z=0;\left(0,\frac{1}{3},\frac{2}{3}\right)$

at $x=0;\left(0,\frac{1}{3},\frac{2}{3}\right)$

similarly we find points G, D and H, E

Required Area $=\frac{\sqrt{3}}{2}-\left[\frac{\sqrt{3}}{18}+\frac{\sqrt{3}}{72}+\frac{\sqrt{3}}{8}\right]$

8. Conceptual

9. $\sum_{r=1}^{\infty}\left(\frac{1}{r^2+r}+\frac{r}{r^4+r^2+1}\right)$

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10. Let $Z = 2e^{i\theta}$

$Z^2 = 4e^{i2\theta}$

$Z^3 = 8e^{i3\theta}$

$\therefore$ area of quadrilateral $= \frac{1}{2}|Z||Z^2|\sin\theta + \frac{1}{2}|Z^2||Z^3|\sin\theta$

$15 = \frac{1}{2}8\sin\theta + \frac{1}{2}32\sin\theta$

$15 = 20\sin\theta$

$\sin\theta = \frac{3}{4}$

$\therefore \text{Im}(Z) = \frac{3}{2}$

11. Let $f(x) = a(x-2)(x-3)(x-4) + x^2$.

Let equation of line AB is $y = bx + C$....

$\therefore$ $f(x) - bx - c$ is cubic where roots are 2, 3, x, where x, is $x-$coordinate of D.

Similarly for AC and BC.

$\therefore$ $\frac{9a-1}{a} = 2+3+x_1 = 3+4+x_2 = 2+4+x_3$.

Adding three equation, we get

$3\left(\frac{9a-1}{a}\right) = 18 + x_1 + x_2 + x_3 = 18 + 24$

$\therefore a = \frac{-1}{5}$

$\therefore f(0) = \frac{-1}{5}(-2)(-3)(-4) = 4.8$

12. Let $\vec{x} = a\hat{i} + b\hat{j} + C$

$\vec{y} = p\hat{i} + q\hat{j} + rk$

$ap + bq + cr = \sqrt{a^2+b^2+c^2}.\sqrt{p^2+q^2+r^2}$

$\therefore \cos\alpha = -1$

$\therefore \vec{x} = \lambda\vec{y}$

13. Conceptual

14. Conceptual

15. $\alpha, \beta, \gamma = 1, 2, -3$.

P) $3! \times 2 = 12$

Q) $3! = 6$

R) $|m| = 0$, $\therefore$ Infinite solution

S) $C_i \to C_1 + C_2 + C_3$ $\therefore |m| = 0$

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16. $Q\left(\frac{5}{4},-1\right), R\left(Q,\frac{3}{2}\right)$

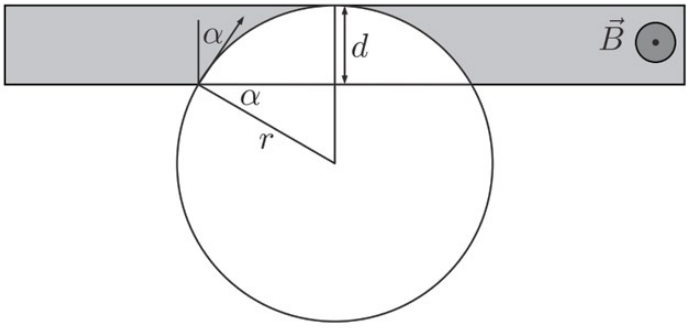
ΔPQR is right angled triangle.

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PHYSICS

17. At limiting case (just grazing the upper boundary):



$$r\sin\alpha_{max} + d = r$$

$$\Rightarrow \sin\alpha_{max} = 1 - \frac{d}{r}$$

$$\alpha_{max} = \sin^{-1}\left(1 - \frac{d}{r}\right)$$

with $r = \frac{mv}{qB}$.

18. Take moments about the point of contact P.
Normal reaction and friction pass through P, so they produce no torque.
Torque of weight about P:

$$\tau_g = mg\left(r\sin\alpha\right).$$

To minimize F, apply it at the point diametrically opposite to P, giving maximum moment arm $2r$.
Torque balance:

$$F_{min}\left(2r\right) = mg\left(r\sin\alpha\right)$$

$$F_{min} = \frac{mg\sin\alpha}{2}.$$

19. $h_{max} = \frac{H + 2H}{2}$

20. The particle leaves the cyclotron when its trajectory radius gets as big as the cyclotron's radius R, in this case its velocity is $v = qBR/m$. During one full circle the particle gets a kinetic energy from $E_k = 2qEd$ the electric field, because the particle goes through the strip two times during this time. Therefore the velocity of the particle when exiting the cyclotron is

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$$\frac{mv^2}{2} = 2qEdn,\ v^2 = \frac{4qEdn}{m}$$

Expressing n from these equations we get $n = \frac{qB^2R^2}{4mEd}$.

21. Initial equilibrium:

$$Mg = kx_0 \Rightarrow x_0 = \frac{Mg}{k}$$

Oscillation period:

$$T = 2\pi\sqrt{\frac{M}{k}} \Rightarrow \tau = \pi\sqrt{\frac{M}{k}} = \frac{T}{2}$$

Electric force shifts equilibrium by:

$$\Delta x = \frac{QE}{k}$$

Each time the field direction changes (every τ), equilibrium shifts by $2\Delta x$, increasing oscillation amplitude by $2\Delta x$.

Given:

$$t = 7\pi\sqrt{\frac{M}{k}} = 7\tau$$

After each interval τ, amplitude increases by $2\Delta x$.
Total displacement after 7τ :

$$x = 14\Delta x = \frac{14QE}{k}$$

22. Total volume:

$$V = \pi R^2 h$$

Initial liquid volume:

$$V_L = kV$$

Initial air volume:

$$V_{a_1} = V - V_L = V(1-k)$$

Mass of liquid:

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$$m = \rho_0 V_L = \rho_0 kV$$

After freezing, density becomes $\lambda\rho_0$, so solid volume:

$$V_S = \frac{m}{\lambda\rho_0} = \frac{kV}{\lambda}$$

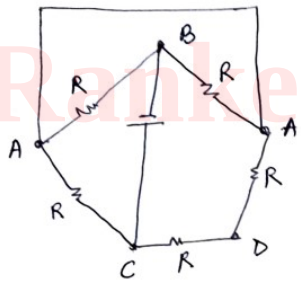
Final air volume:

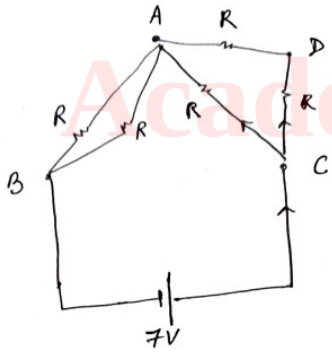
$$V_{a2} = V - V_S = V\left(1 - \frac{k}{\lambda}\right)$$

Using ideal gas law:

$$\frac{P_1 V_{a_1}}{T_1} = \frac{P_2 V_{a_2}}{T_2}$$

$$\frac{P_2}{P_1} = \frac{(1-k)}{\left(1-\frac{k}{\lambda}\right)} \cdot \frac{T_2}{T_1}$$





23.

$$\text{Req} = \frac{R}{2} + \frac{2R}{3}$$

$$\text{Req} = \frac{7R}{6}$$

- Voltmeter reads $|V_B - V_D|$
- $\text{Req} = \frac{7}{6}\Omega \quad (R = 1\Omega)$
- $i_{\text{battery}} = \frac{7}{7/6} = 6\text{A}$

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- If $V_B = 0 \qquad V_D = 7 - i_{CD}R$

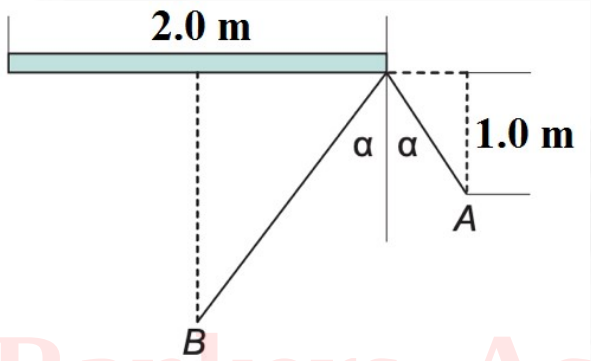
$$i_{CD} = \frac{1}{3} i_{net} = 2A$$

$$V_D = 7 - 2 \times 1 = 5V$$

24. The ball divides the tube of the cannon into two chambers. Before firing the pressure in both chambers is equal to zero. By rupturing the membrane on the left the tube's left side fills with outer air and between each side of the ball there is pressure difference P_0.

$$v = \sqrt{2la} = \sqrt{\frac{2lP_0\pi d^2}{4m}} = d\sqrt{\frac{lP_0\pi}{2m}}$$

25.



At time 't'

- Horizontal distance of A from mirror edge:

$$(2 - t)$$

- Vertical distance of B from mirror:

$$(3.5 - t)$$

For visibility, line joining B and image of A just touches mirror edge $\Rightarrow$ similar triangles:

$$\frac{1}{3.5 - t} = \frac{2 - t}{1}$$

Rearranging:

$$(3.5 - t)(2 - t) = 1$$

$$t^2 - 5.5t + 6 = 0$$

$$t = 1.5 \text{ s}, 4.0 \text{ s}$$

Since earliest time is required,

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$$t = 1.5 \text{ s}$$

26. Let the road be along unit vector i.
Let wind velocity (ground frame) be

$$\vec{w} = u\hat{i} + \vec{p}$$

where u is the component along the road and $\vec{p}$ is perpendicular to it.
Cyclist velocities:

$$\vec{v}_1 = 10\hat{i},\ \vec{v}_2 = -20\hat{i}$$

Given relative wind speeds:

$$|\vec{w} - \vec{v}_1| = 20,\ |\vec{w} - \vec{v}_2| = 20$$

So

$$(u-10)^2 + p^2 = 20^2 = 400$$

$$(u+20)^2 + p^2 = 400$$

Subtract (1) from (2):

$$(u+20)^2 - (u-10)^2 = 0 \Rightarrow 60u + 300 = 0 \Rightarrow u = -5 \text{ km h}^{-1}$$

Insert in (1):

$$(-5-10)^2 + p^2 = 400 \Rightarrow 225 + p^2 = 400 \Rightarrow p^2 = 175$$

Hence true wind speed:

$$|\vec{w}| = \sqrt{u^2 + p^2} = \sqrt{25+175} = \sqrt{200}\ \frac{\text{km}}{\text{hr}}$$

27. Case-1: Pressed at $\frac{3}{7}L$

The vibrating segment has length:

$$L_1 = \frac{3}{7}L$$

For a stretched string,

$$f_1 = \frac{v}{2L_1} = \frac{v}{2\left(\frac{3}{7}L\right)} = \frac{7v}{6L}$$

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Case-2: Lightly touched at $\frac{3}{7}L$

Touching $\rightarrow$ forces a node at that point.

Thus, the whole string vibrates but with a node at $\frac{3}{7}L$

Both segments must contain integer number of half-wavelengths.
Segment lengths:

Their ratio: $\frac{3}{7}L$ and $\frac{4}{7}L$

Their ratio: $\frac{\frac{3}{7}L}{\frac{4}{7}L}=\frac{3}{4}$

So admissible mode numbers are **3** and **4** half-wavelengths, respectively.

Thus wavelength of the allowed standing wave:

$$\lambda=\frac{2L}{7}$$

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Frequency in Case-2:

$$f_2=\frac{v}{\lambda}=\frac{v}{\frac{2L}{7}}=\frac{7v}{2L}$$

Ratio

$$\frac{f_2}{f_1}=\frac{\frac{7v}{2L}}{\frac{7v}{6L}}=\frac{6}{2}=3$$

28. After long OFF, capacitor is fully discharged, so $V_R=9\text{V}$:

$$\frac{9^2}{R}\le 0.25\Rightarrow R\ge 324\Omega$$

If polarity is reversed without turning OFF, capacitor holds 9V, so total across R:

$$V_R=18\text{ V}$$

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$$\frac{18^2}{R} > 0.25 \Rightarrow R < 1296\Omega$$

$$324\Omega < R < 1296\Omega$$

29. For the given instant, the perpendicular distance of the line of motion of C from A is l and $AC = 2l$, angle between AC and $\vec{v}$ is θ,

$\Rightarrow \sin\theta = \frac{l}{AC} = \frac{l}{2l} = \frac{1}{2} \Rightarrow \theta = \angle(AC, \vec{v}) = 30^\circ.$

Since $AB = BC = AC = 2l$, ΔABC is equilateral, so $\angle BAC = 60^\circ$.

Hence the angle between AB and the velocity direction is 90°, i.e. $AB \perp \vec{v}$

Because rod BC is rigid, the components of the velocities of B and C along BC must be equal.
The angle each velocity makes with BC is the same, so

$v_B = v_C = v.$

Thus, at that instant, point B executes circular motion about A with speed v and radius AB = $2l$. The component of its acceleration along AB is therefore the centripetal one:

$$a_{B\|AB} = \frac{v^2}{2l}.$$

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Now shift to a frame moving with velocity $\vec{v}$ (same as the instantaneous velocity of B and C).
In this frame, at that instant, B is momentarily at rest, so it has no centripetal acceleration about C; it acceleration is purely tangential and hence perpendicular to BC.

This acceleration vector makes an angle of 30° with AB, so its projection on AB equals

$$\frac{v^2}{2l} : a\cos 30^\circ = \frac{v^2}{2l} \Rightarrow a = \frac{v^2}{2l\cos 30^\circ} = \frac{v^2}{\sqrt{3}l}.$$

30. Final pressure of both the part will be same and the gas in the right part of the piston follow adiabatic process.

31. Using ampere's law we get

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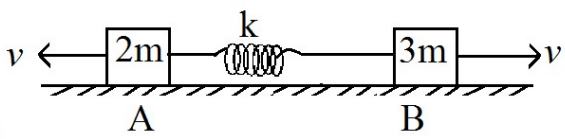
$$B=\begin{cases}\mu_0 j_0\left(\dfrac{r}{2}-\dfrac{r^2}{3R}\right) & r\le R\\ \dfrac{1}{6}\mu_0 j_0 R & r=R\\ \dfrac{\mu_0 j_0 R^2}{2r} & r\ge R\end{cases}$$

At $r=\dfrac{R}{2}\Rightarrow B=\dfrac{1}{6}\mu_0 j_0 R$

At $r=\dfrac{3R}{4}\Rightarrow B=\dfrac{3}{16}\mu_0 j_0 R$

At $r=2R\Rightarrow B=\dfrac{1}{12}\mu_0 j_0 R$

32.



$$V_{com}=\frac{3mv-2mv}{5}=v/5$$

- In frame of com

$$\vec{V}_{ACOM}=\frac{-6v}{5} \text{ and } \vec{V}_{B_{COM}}=\frac{4v}{5}$$

- Both perform SHM in frame of com

A) $\vec{V}_A=\vec{V}_{Acom}+\vec{V}_{com}=\vec{V}_{Acom}+\dfrac{v}{5}\hat{i}$

when $\vec{V}_{Acom}=-\dfrac{v}{5}\hat{i}$ $\boxed{\vec{V}_{A\,\min}=0}$

B) For $\vec{V}_{A\,\max}=\dfrac{6v}{5}+\dfrac{v}{5}=\dfrac{7v}{5}$ (when A moves to right at $6/5\,\vec{V}_A$ will be max)

C) For $\vec{V}_{B_{\max}}=\vec{V}_{B_{(\max)}}+\vec{V}_{com}=\dfrac{4V}{5}+\dfrac{V}{5}=V$

D) $V_{com}=\dfrac{3mv-2mv}{5m}=v/5$

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33. Spodumene is a lithium aluminium silicate mineral having chemical formula $LiAl(SiO_3)_2$. Rest of the options have 'lith' in their name but don't contain lithium as a constituent element.

34. Both white and red phosphorus are poor electrical conductors. Black phosphorus, thermodynamically most stable allotrope of phosphorus is a good electrical conductor.

35. a) $60\mathbf{OH}^- + 160\mathbf{H}^+ \rightarrow 60\mathbf{H_2O}$

b) $40\mathbf{OH}^- + 240\mathbf{H}^+ \rightarrow 40\mathbf{H_2O}$

c) $80\mathbf{OH}^- + 80\mathbf{H}^+ \rightarrow 80\mathbf{H_2O}$

d) $50\mathbf{OH}^- + 200\mathbf{H}^+ \rightarrow 50\mathbf{H_2O}$

Since millimoles of H_2O formed is maximum in option-(C), it will cause maximum evolution of heat.

36. $PbCl_2 \rightleftharpoons \underset{S}{Pb^{+2}} + \underset{2S}{2Cl^-}$

$K_{sp}(PbCl_2) = 1.6\times10^{-5}$

$4\,S^3 = 1.6\times10^{-5}$

$4\,S^3 = 16\times10^{-6}$

$S^3 = 4\times10^{-6}$

$S = \sqrt[3]{4}\times10^{-2}$

The cube root of 4 will lie between 1.5 and 1.6 (by inspection)

Taking $S = 1.5\times10^{-2}$ moles/lit (lesser value)

$[Cl^-] = 2S = 3\times10^{-2}$ mole/lit

$= 0.03\times143.5$ g/lit

$= 4.305$ g/lit

$= 0.4305$ g/100ml

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So in option-D even if **all** $PbCl_2$ were to dissolve, the chloride ion concentration will be greater in option-C as more weight of $PbCl_2$ shall dissolve here.

37. Dimethyl (not diethyl) amine will attack in step-3

38. Option - D is wrong because that would mean equal weights of Hg and I_2 are reacting

39. a) $S_\alpha \rightleftharpoons S_\beta$ (transition temperature is 369 K or $96^\circ C$)

b) $Sn_\beta \rightleftharpoons Sn_\alpha$ (transition temperature is 286 K or $13^\circ C$)

c) α-glucose(36%) $\rightleftharpoons$ β-glucose(64%)

d) ${}_{92}U^{238} \rightarrow {}_{82}Pb^{206} + \mathbf{8}_2He^4 + \mathbf{6}_{-1}e^0$

40. $K, L, M, N, O \ldots$ (i.e $n = 5$)

atom 1	$5 \longrightarrow 4 \longrightarrow 3 \longrightarrow 2 \longrightarrow 1$	OR	$5 \longrightarrow 4 \longrightarrow 3 \longrightarrow 2 \longrightarrow 1$
atom 2	$5 \longrightarrow 3 \longrightarrow 1$		$5 \longrightarrow 3 \longrightarrow 1$
atom 3	$5 \longrightarrow 4 \longrightarrow 1$		$5 \longrightarrow 2 \longrightarrow 1$

In either cases a maximum of 7 **distinct** spectral lines will be observed.

41. $PV = nRT$

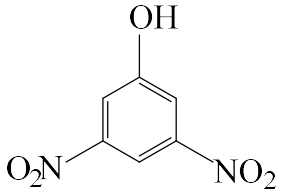
$$\frac{748}{760} \times 75 = n \times 0.0821 \times 298$$

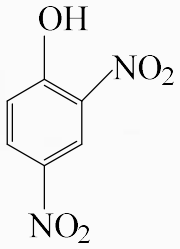
$$n = 3.017$$

$$\text{moles of } NaN_3(s) = \frac{6}{9} \times 3.017 = 2.011$$

$$\text{mass of } NaN_3(s) = 2.011 \times 65 = 130.71 \text{ g}$$

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42. Least acidic dinitrophenol:  $(pk_a = 6.7)$

Most acidic dinitrophenol:  $(pk_a = 4.1)$

43. Serial nos. 1 and 4 are chiral molecules and serial nos. 1,2 and 3 are dicarboxylic acids.

44. $Y\,Ba_2\,Cu_3\,O_7$

45.

	HA	$\rightleftharpoons$	H^+	A^-
	c		0	O
	$c(1-\alpha)$		$c\alpha$	$c\alpha$

$$K_a = \frac{c^2\alpha^2}{c(1-\alpha)} = \frac{c\alpha^2}{1-\alpha} = c\alpha^2 \text{ (by neglecting } \alpha \text{ w.r.t 1)}$$

$$\alpha = \sqrt{\frac{Ka}{c}} = \sqrt{\frac{2\times10^{-4}}{0.5}} = 2\times10^{-2} = 0.02$$

$$\left[H^+\right] = c\alpha = 0.5\times0.02 = 10^{-2}\,M; pH = 2$$

	HA	$\rightleftharpoons$	H^+	+	A^-
	$\frac{0.5}{3333}$ or $\frac{1}{6666}$ M		0		0
			$\frac{1}{6666}\alpha$		$\frac{1}{6666}\alpha$

$$\frac{1}{6666}(1-\alpha)$$

New $pH = 4$ (double of previous value)

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$$\frac{1}{6666}\alpha = 10^{-4}$$

$$\alpha = 0.6666$$

46. Partial forced pairing will occur in $\left[Cr(en)_3\right]^{2+}$ due to which 2 unpaired electrons will remain in t_{2g} orbital while the e_g orbital will get vacated.

47. $C(s) + \frac{1}{2}O_2(g) \longrightarrow CO(g)$

Since both the number and complexity of gas molecules increase towards the product side, ΔS_f^0 must be greater than zero. Since IV is only being matched with either S or T among the options, one should pick T only for IV.
However since $I \to P$ is the simplest of all matchings, its clear that IV MUST be matched with T only. Formation of O_3 gas (Unit 7, 7.13, page 186, NCERT ClassXII) is endothermic and so is the formation of $I_2(g)$ from $I_2(s)$

48. α -carbonyl halide is more reactive than benzyl halide towards S_{N_2} reaction.

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